Customer Retail Machine Learning Project

Dataset Output

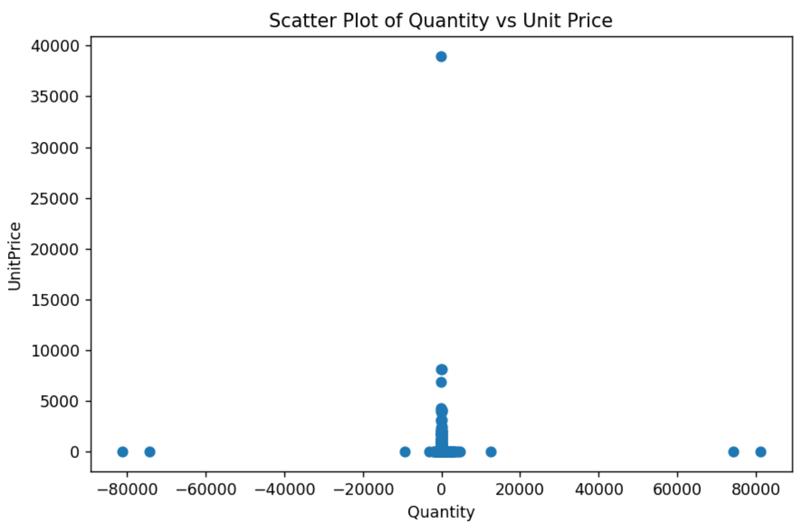
```
df = pd.read_csv("customer_retail csv file.csv")

print(df.head())
```

```
  InvoiceNo StockCode                          Description  ... UnitPrice CustomerID         Country
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER  ...      2.55    17850.0  United Kingdom
1    536365     71053                  WHITE METAL LANTERN  ...      3.39    17850.0  United Kingdom
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER  ...      2.75    17850.0  United Kingdom
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE  ...      3.39    17850.0  United Kingdom
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.  ...      3.39    17850.0  United Kingdom

[5 rows x 8 columns]
```

Scatter Plot



Logistic Regression

```
print("LOGISTIC REGRESSION")

log_model = LogisticRegression()

log_model.fit(x_train, y_train)

y_pred_log = log_model.predict(x_test)

log_accuracy = accuracy_score(y_test, y_pred_log)

print("Accuracy:", log_accuracy)

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred_log))
```

```
Accuracy: 0.8898065531057199
Confusion Matrix:
[[    0     0     0 ...     0   229     0]
 [    0     0     0 ...     0    78     0]
 [    0     0     0 ...     0     5     0]
 ...
 [    0     0     0 ...     0    13     0]
 [    0     0     0 ...     0 72399     0]
 [    0     0     0 ...     0    43     0]]
```

Decision Tree

```
print("\nDECISION TREE")

dt_model = DecisionTreeClassifier()

dt_model.fit(x_train, y_train)

y_pred_dt = dt_model.predict(x_test)

dt_accuracy = accuracy_score(y_test, y_pred_dt)

print("Accuracy:", dt_accuracy)

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred_dt))
```

```
DECISION TREE
Accuracy: 0.8903104490819261
Confusion Matrix:
[[   13     0     0 ...     0   197     0]
 [    0     0     0 ...     0    75     0]
 [    0     0     0 ...     0     5     0]
 ...
 [    0     0     0 ...     0    13     0]
 [   22     0     0 ...     1 72218     0]
 [    0     0     0 ...     0    43     0]]
```

KNN

```
print("\nKNN")

knn_model = KNeighborsClassifier()

knn_model.fit(x_train, y_train)

y_pred_knn = knn_model.predict(x_test)

knn_accuracy = accuracy_score(y_test, y_pred_knn)

print("Accuracy:", knn_accuracy)

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred_knn))
```

```
KNN
Accuracy: 0.8851977484452965
Confusion Matrix:
[[   18     0     0 ...     0   180     0]
 [    0     1     0 ...     0    75     0]
 [    0     0     0 ...     0     5     0]
 ...
 [    0     0     0 ...     0    12     0]
 [   36     1     0 ...     0 71772     0]
 [    0     0     0 ...     0    43     0]]
```

Accuracy Comparison Graph